

Tutorial: Why RETURN LOSS > -10dB And VSWR < 2 FOR ANTENNA DESIGN

Return loss of signal and voltage standing wave ratio (VSWR) are two of the most important characteristics of an antenna. So their proper understanding is essential for a good antenna



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Antenna plays a major role at present in all fields of wireless communication. Return loss and VSWR are the two major indices to measure the input characteristics performance of good antenna and any RF circuits in general as a network, which can be measured using a network analyser.

Return loss is the parameter that indicates the portion of power rejected and returned to any source that intended to transmit it. It represents the percentage of impedance matching between the two points (source and load) in power transmission in terms of portion of power reflected back to source from load. Higher the impedance matching lesser will be the return loss.

Return loss (R)

Generally, the return loss can be defined as the amount of power returned or reflected due to discontinuity in the path

of transmission or the impedance mismatch, which will also be detailed here. It is the ratio of incident or input power (P_i) to the reflected or return power (P_r) measured in decibel (dB).

Return loss can be related to

reflection coefficient and voltage standing wave ratio (VSWR). Reflection coefficient is the ratio of incident or forward voltage to output or reflected voltage represented by Γ and related to return loss as follows:

$$\text{Return loss (R)} = 10 \log \left(\frac{P_i}{P_r} \right)$$

$$R \text{ (dB)} = -20 \log (\Gamma)$$

Impedance (Z)

Fig. 2 shows the three components involved in radio transmission, that is, the source to transmit radio signals, the transmission line, and the antenna as the load. The figure shows the matched and the mismatched conditions.

This classification is based on the difference between the numbers in the figures indicated near the radio (50Ω) and antenna (75Ω). These numbers represent the impedance (Z) of the concerned device, which is complex in nature and is a combination of resistance (R) and inductive reactance (L) or/and the capacitive reactance (C). Remember that the value of impedance depends on frequency of operation.

Voltage standing wave ratio (VSWR)

VSWR indicates how well the source or load is matched in impedance with the antenna and the transmission line used for power transmission. It represents the maximum and minimum voltage level along the entire length of the transmission line as follows:

$$\text{VSWR} = \frac{V_{\max}}{V_{\min}} = \frac{1+\Gamma}{1-\Gamma}; (\Gamma) = \frac{\text{VSWR}-1}{\text{VSWR}+1}$$

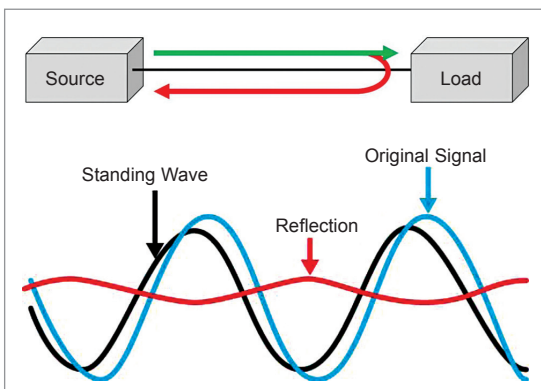


Fig. 1: Depiction of return loss